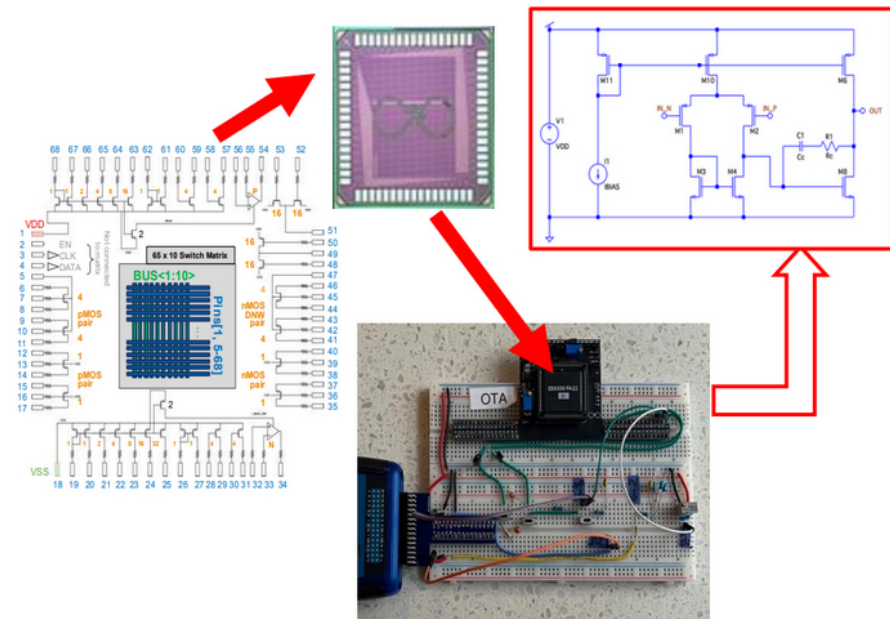


IEEE SSCS Chip-A-Thon 2026



MOSbius Track

- Track Leads: Peter Kinget (peterkinget), Juan Moya Baquero (jsmoya), Luighi (LuighiV)
- Mentors: Andrew Chon (asc), Xianglin Pu (Xianglin Pu)



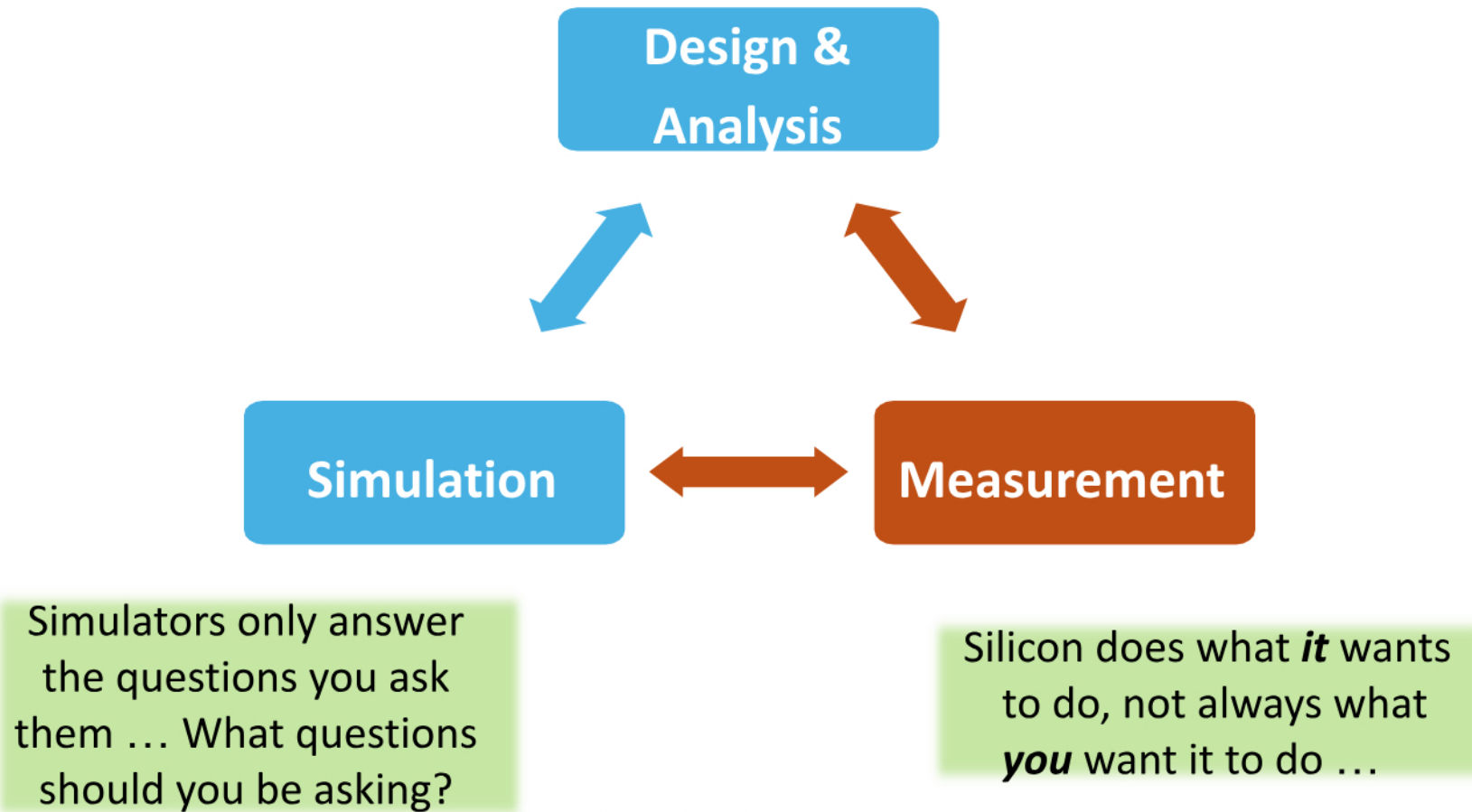
MOSbius Mission



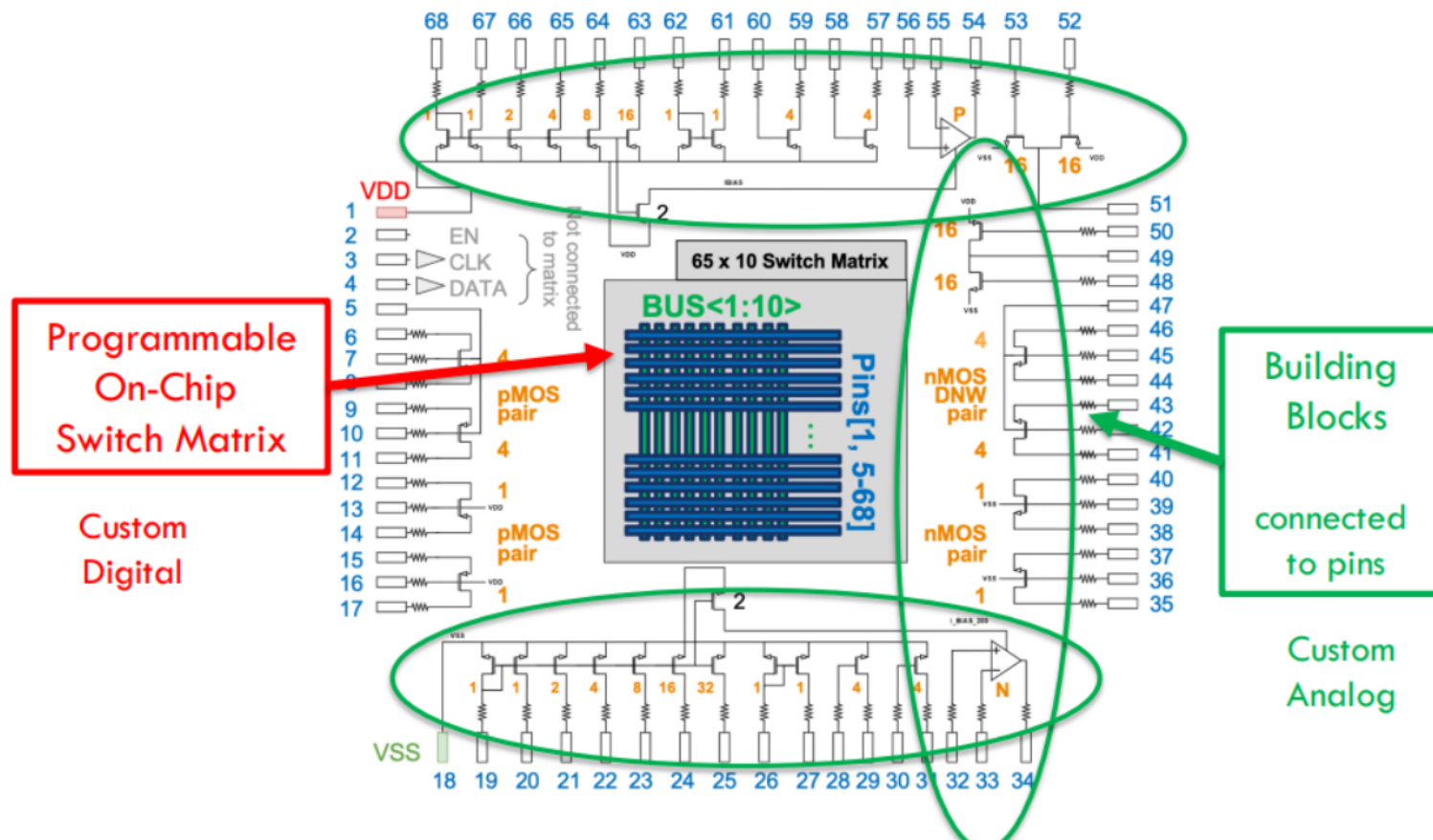
Make it easy for learners to experiment with CMOS transistors in circuits **relevant to IC design**

<https://mosbius.org>

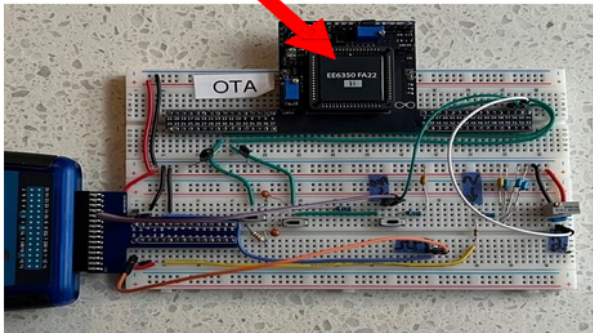
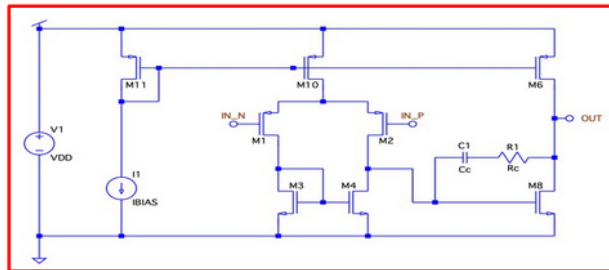
Motivation: Learning IC Design



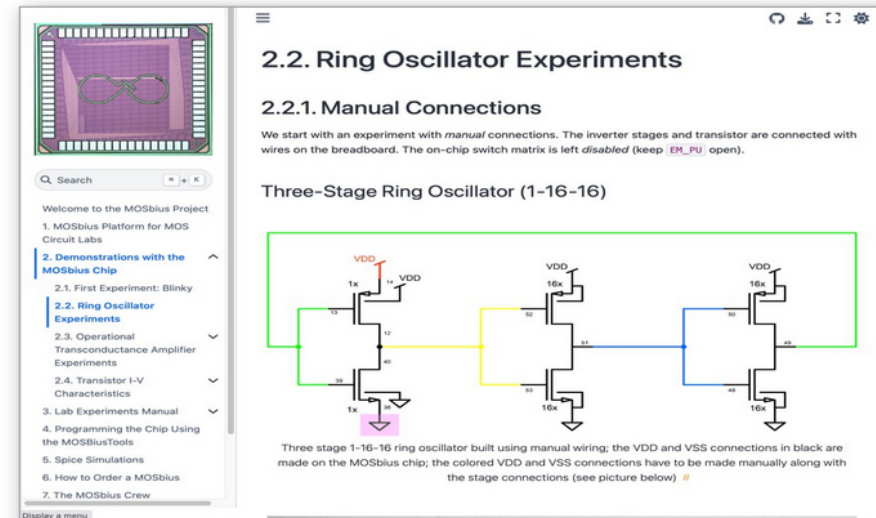
What is MOSbius, so far?



What is MOSbius, so far?



Current version: focus
on transistor-level
analog design



- Website
- Programming support
- Simulation support

<https://mosbius.org>

MOSbius Chipathon Ideas



- Port current MOSbius (in part) to GF180
 - Significant benefits that everything will be in open-source, in particular layouts □ 3D animations, ...
- Develop ‘*digital-transistor-level*’ MOSbius
 - DFFs, logic gates, ... at circuit level
- (very advanced) RF MOSbius
 - RF building blocks to configure various RF receiver topologies
- **Your Ideas**

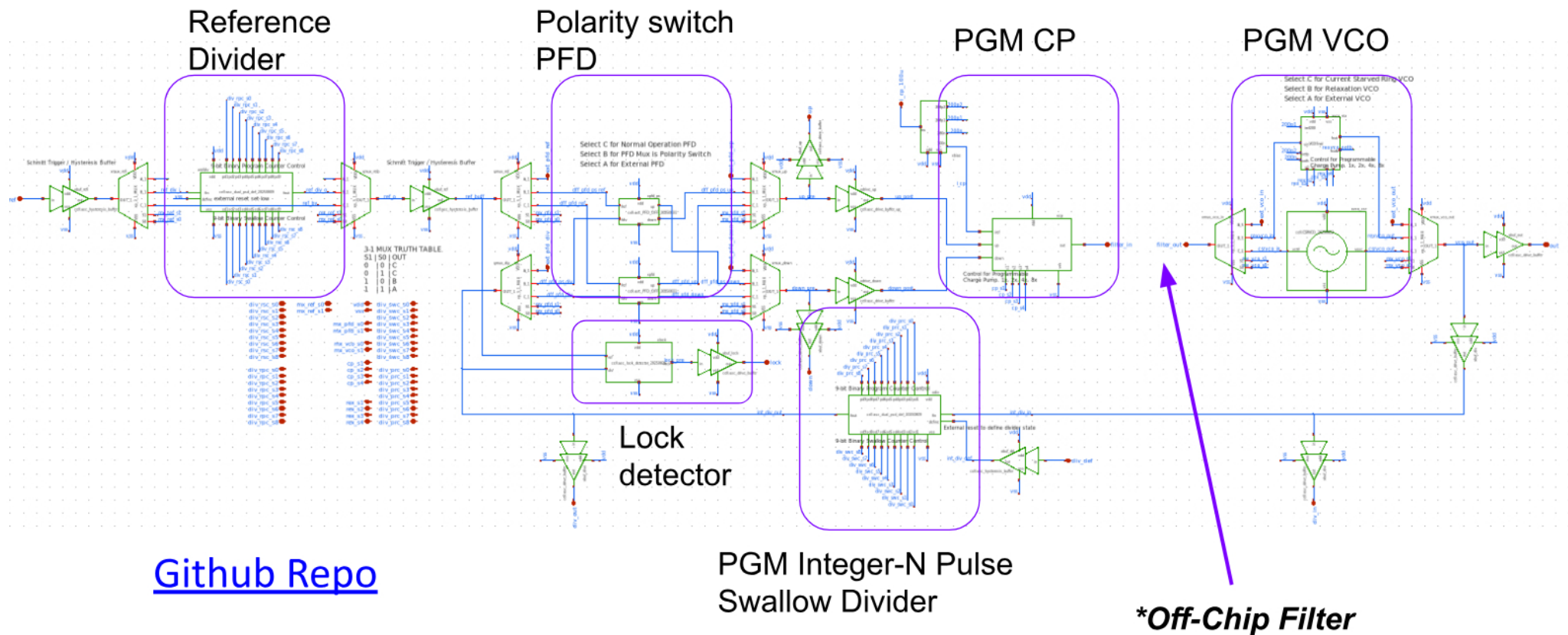
MOSbius Chipathon Goals



- Think about use cases for your chip
 - What experiments can your chip support?
 - How will your chip & experiments draw people into IC design?
 - How easy will your chip be to use, program, ...

The design goals are not only about the design and tape-out, but also about your chip's application, i.e., what experiments your chip can support.

Example: PLL MOSbius



[Github Repo](#)

Example: PLL MOSbius, MOSbius v2

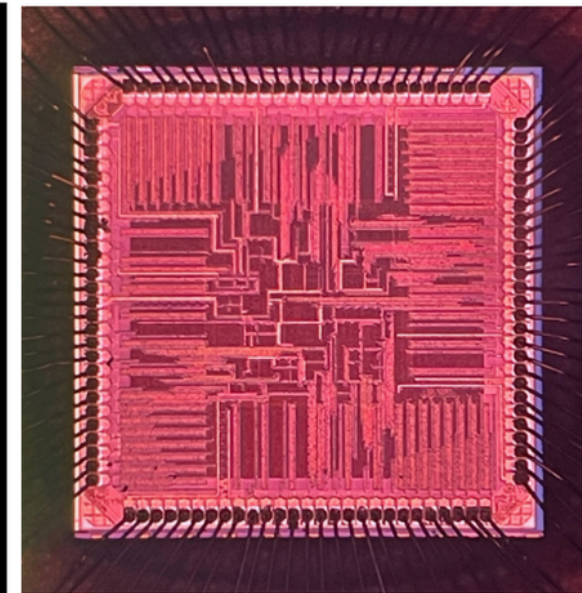
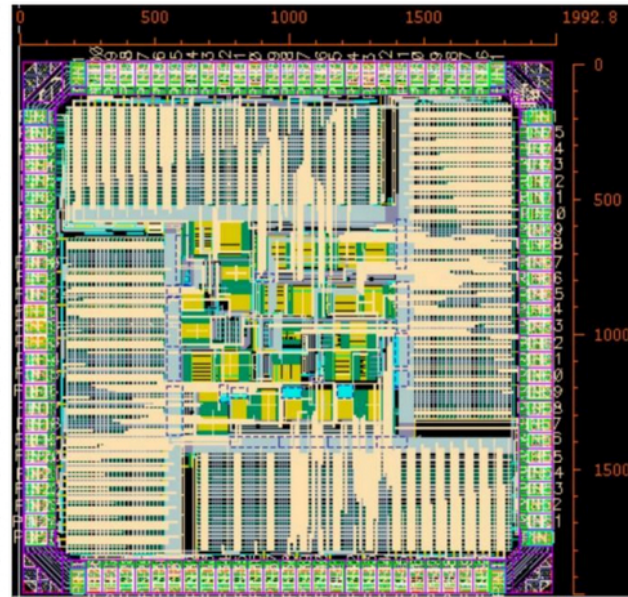
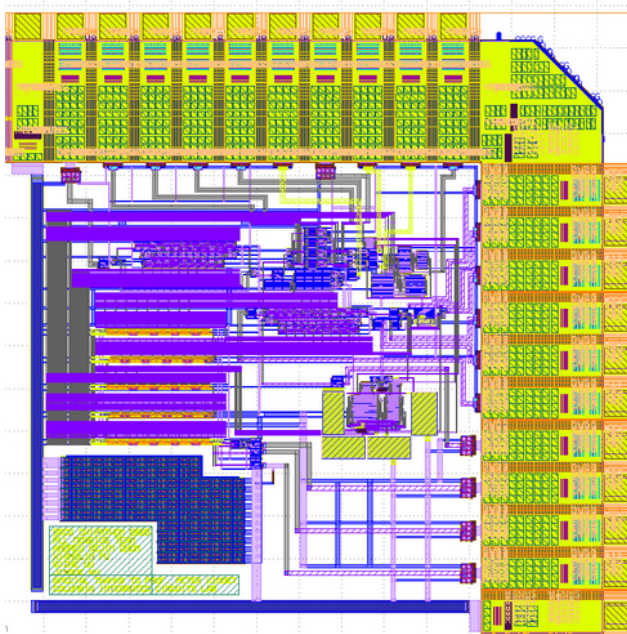
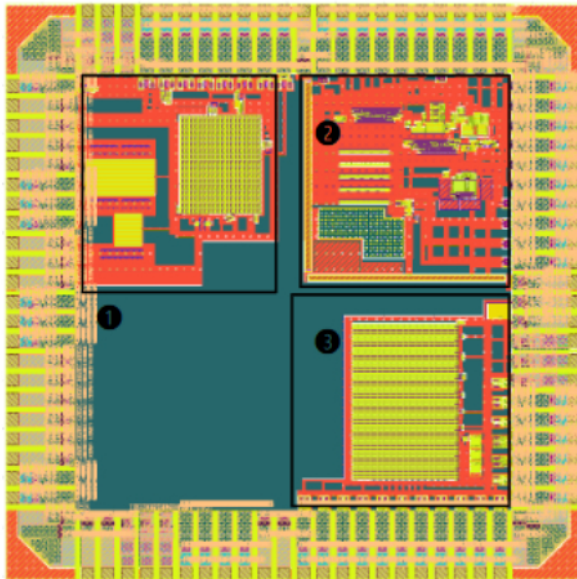


Fig 1. MOSbius V2 TOP Level Layout View (Left), and Die (Right)

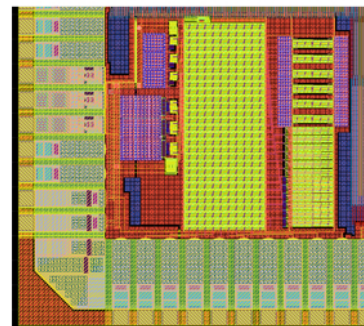
Chipathon 2025 MOSbius Track



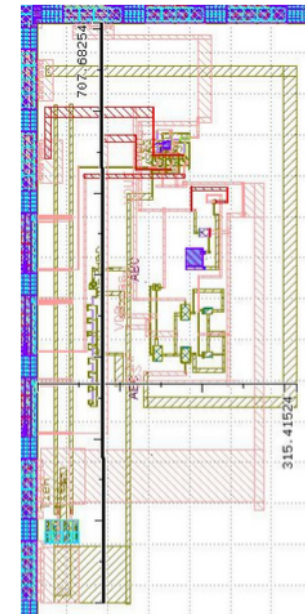
Track M1



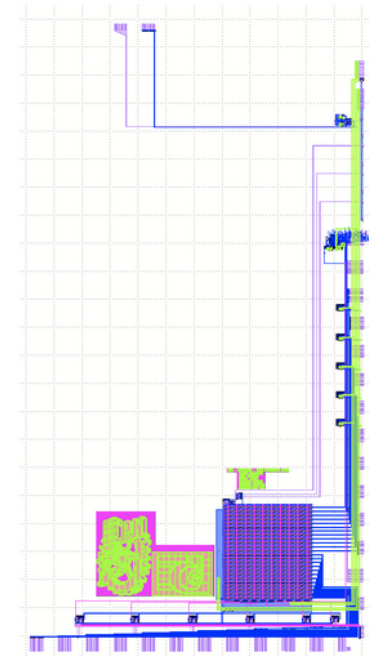
- XDAC
 - MOSbius-style Class D-Amplifier with Programmable Modulation Frequency (<400 kHz)
- RF MOSbius:
 - Programmable type-II PLL (5MHz - 150MHz)
- Low Noise Anomaly:
 - Field Programmable Transistor Array



CreActive - OPAMP with R/C for 100kHz LP Biquad-Filter

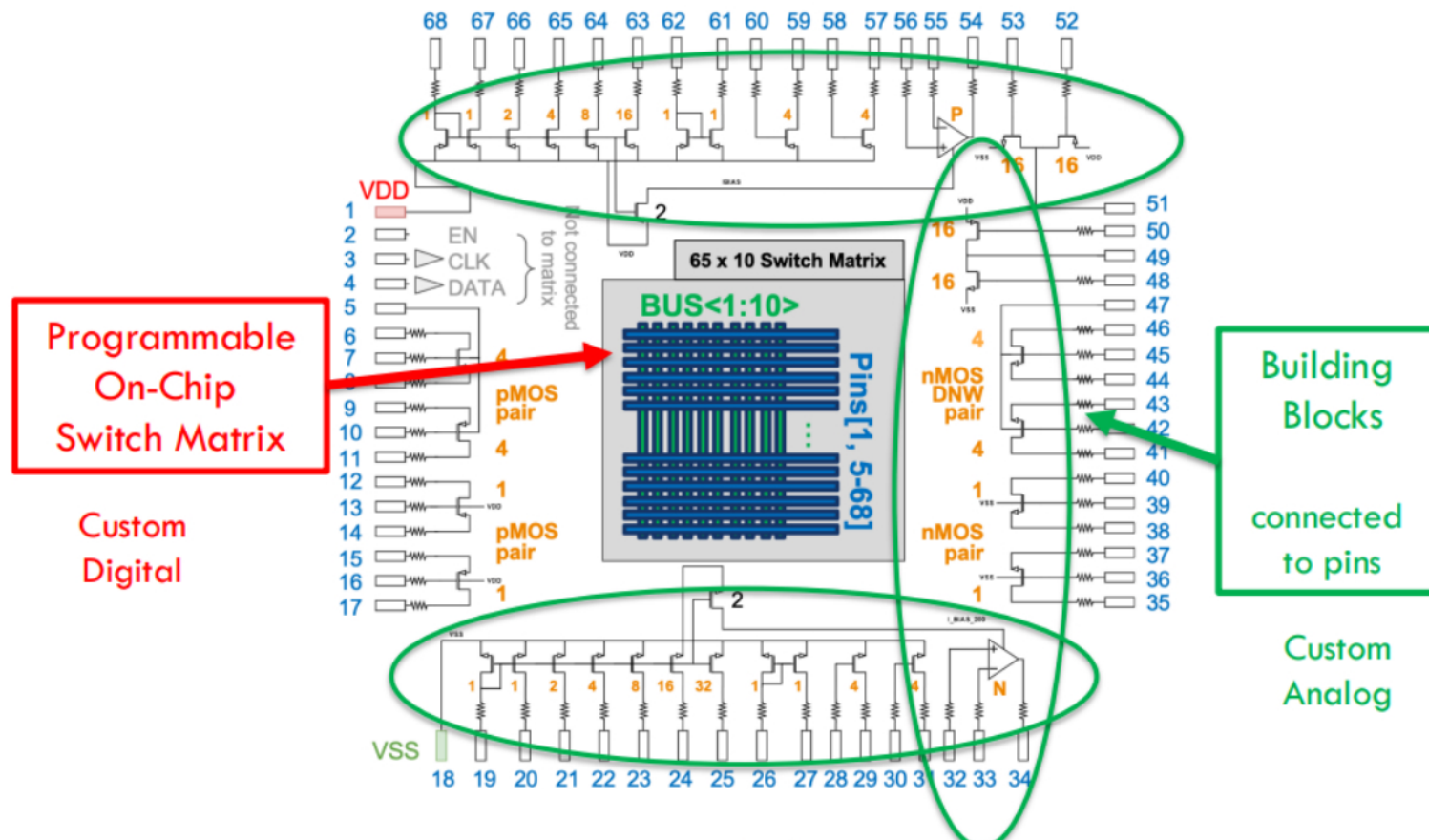


Team RAVAVUS - PGA



SpikeCore - reconfigurable neuromorphic circuit

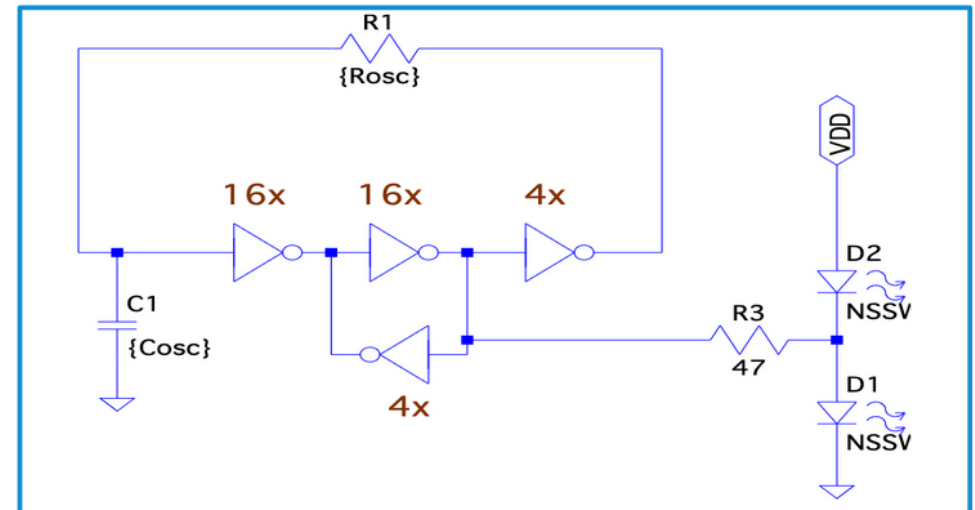
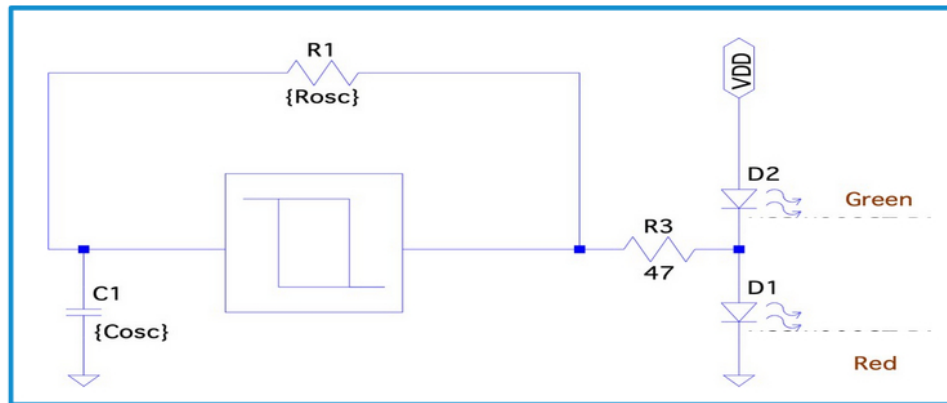
MOSbius Chip



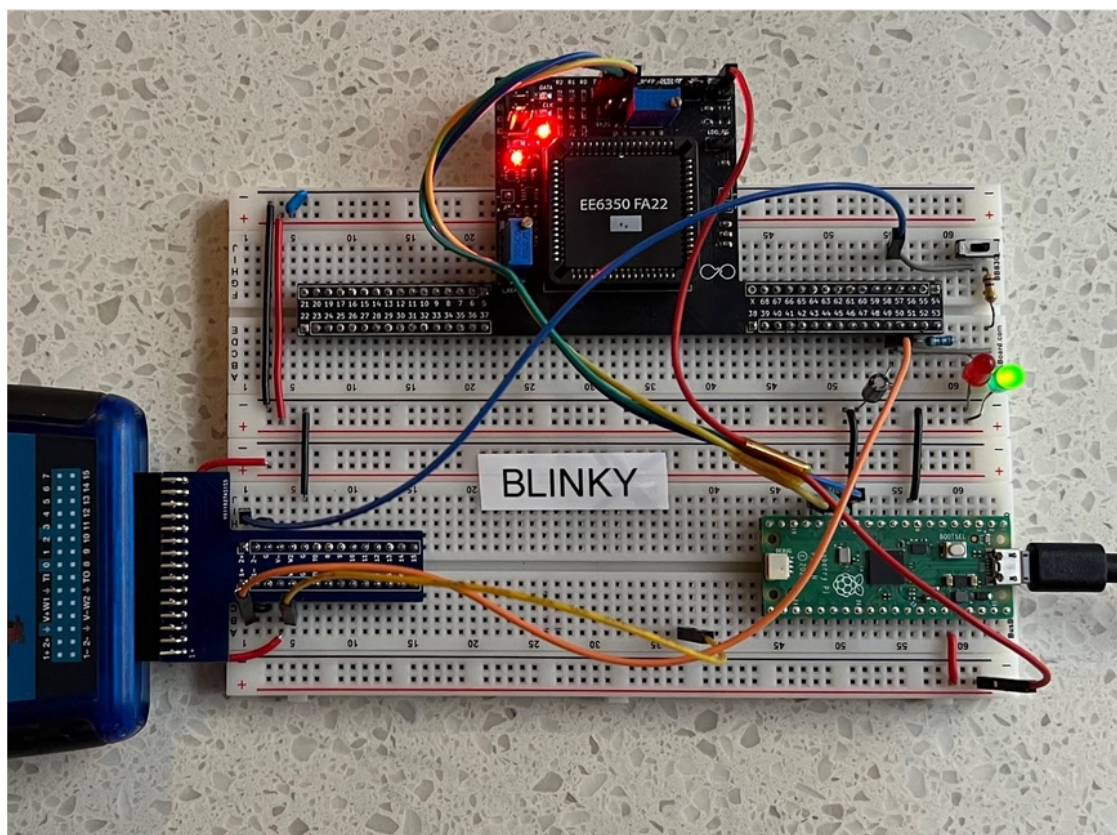
A First Example: Blinky



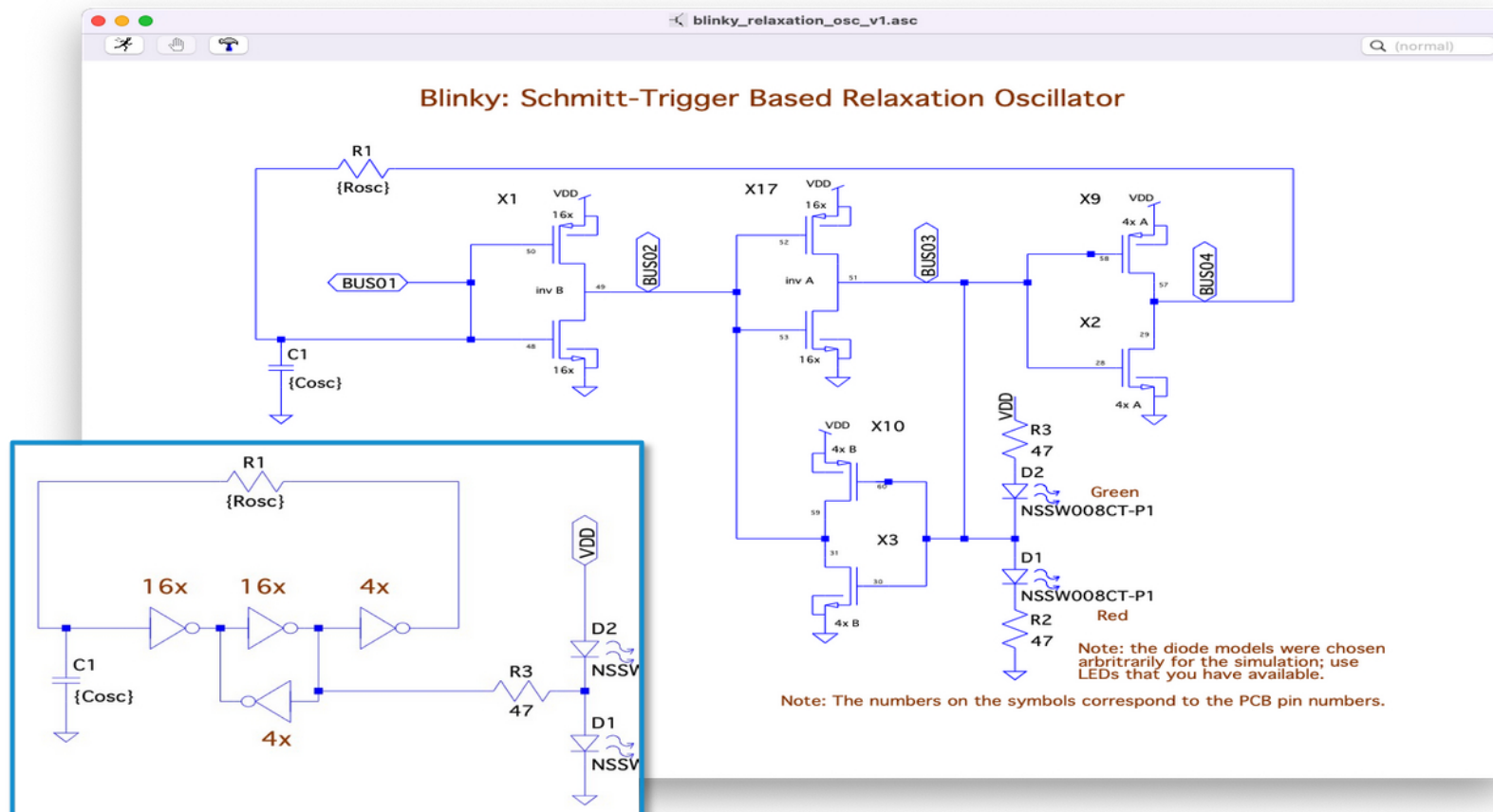
Schmitt-Trigger based Relaxation Oscillator



Positive Feedback $\square$ Hysteresis



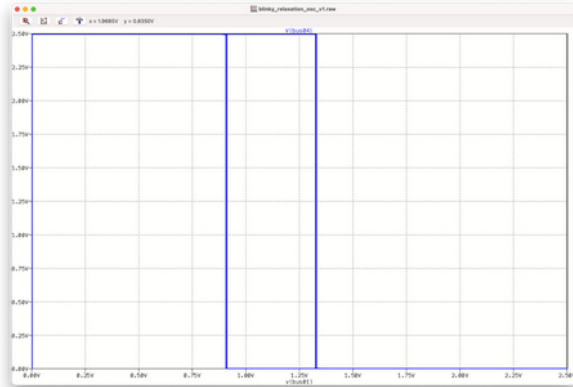
Blinky Spice Schematic



Blinky Schmitt Trigger Simulation

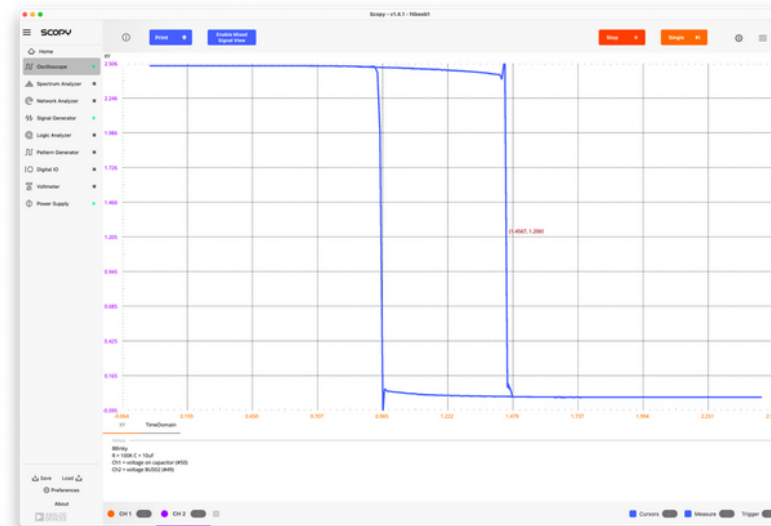


Simulation



$$\Delta V = 418mV$$

Measurement

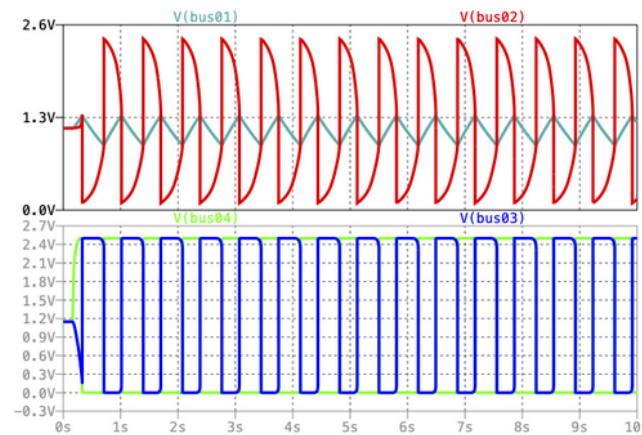


$$\Delta V = 505mV$$

Blinky



Simulation

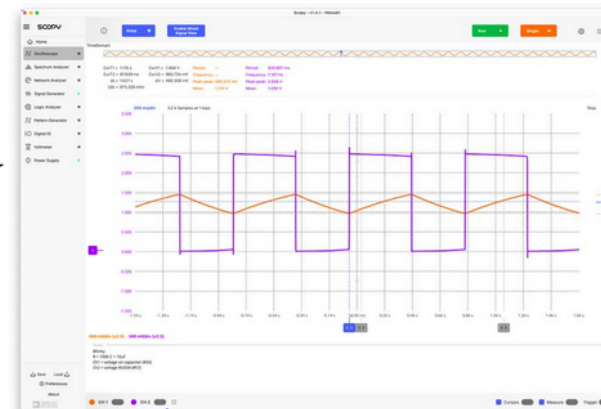


$$\Delta V = 0.411 V$$
$$T = 682 ms$$

Measurement



$$\Delta V = 0.496 V$$
$$T = 837 ms$$



Blinky



$$C \frac{\Delta V}{\Delta T} = I_{ch} \approx \frac{V_{DD} - V(BUS01)_{avg}}{R}$$
$$\Delta T \approx RC \frac{\Delta V}{V_{DD} - V(BUS01)_{avg}}$$
$$\Delta V = 0.5 V \Rightarrow \Delta T \approx 400ms$$
$$T = 2\Delta T = 800ms$$

Design &
Analysis

Note:

$$682 \times \frac{496}{411} = 823$$

$$\Delta V = 0.411 V$$
$$T = 682ms$$

Simulation

Measurement

$$\Delta V = 0.496 V$$
$$T = 837ms$$

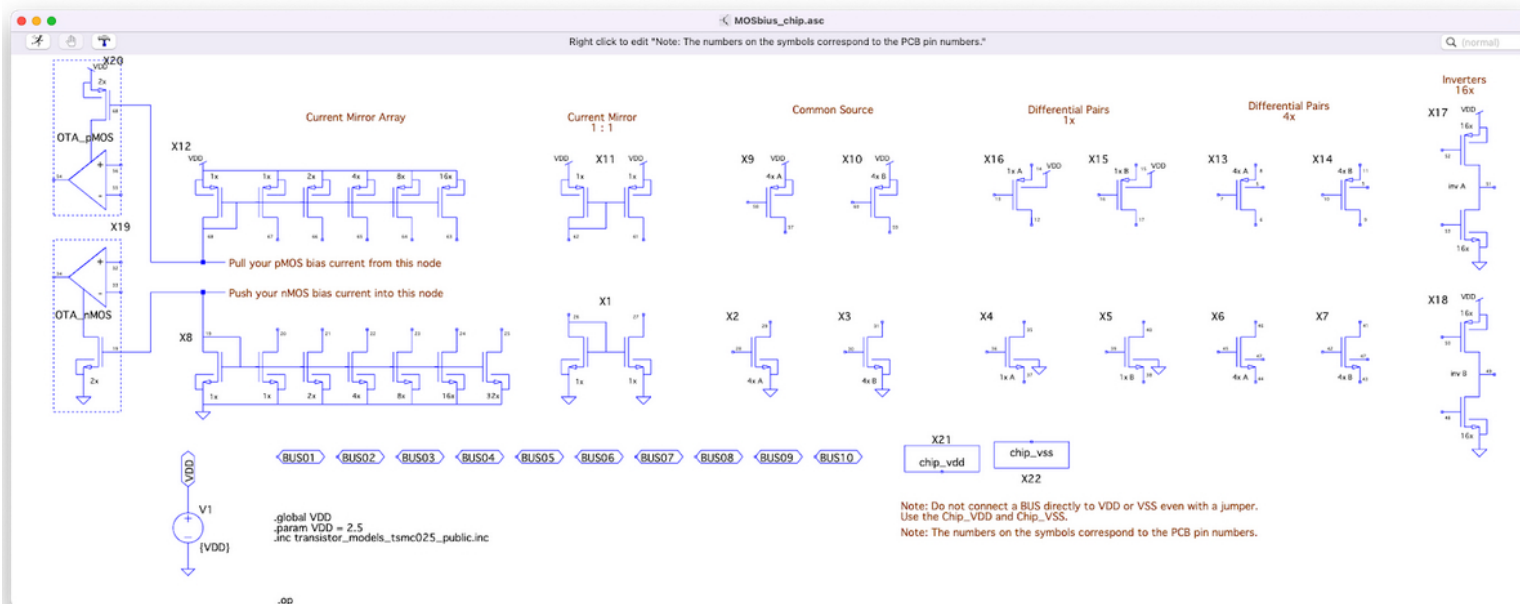
Main difference in ΔV due to device modeling inaccuracies in public model card



Spice Simulations



- Custom LTspice Symbol Library

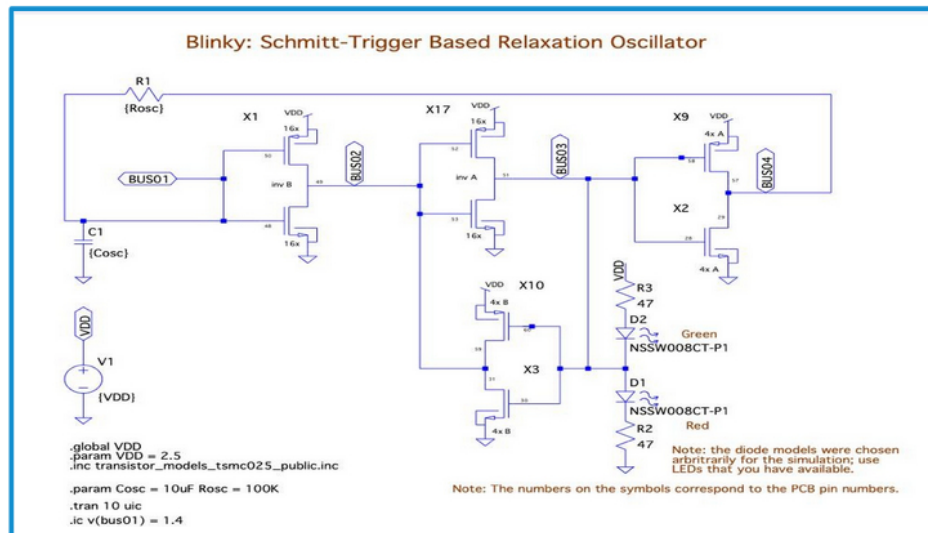


- Public domain BSIM3 transistor model card
- Can easily be ported to ngspice, commercial simulators, ...

MOSbius Tools



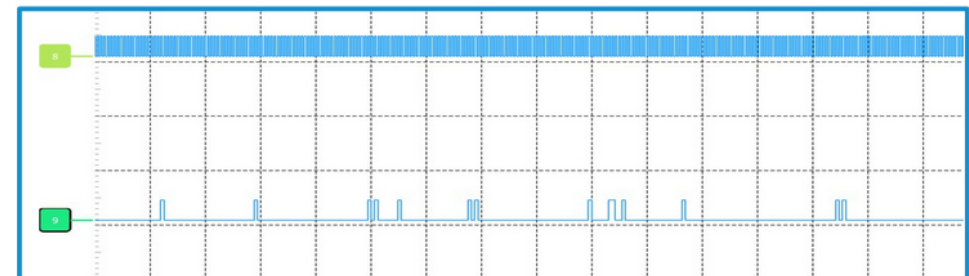
Schematic



connections.json

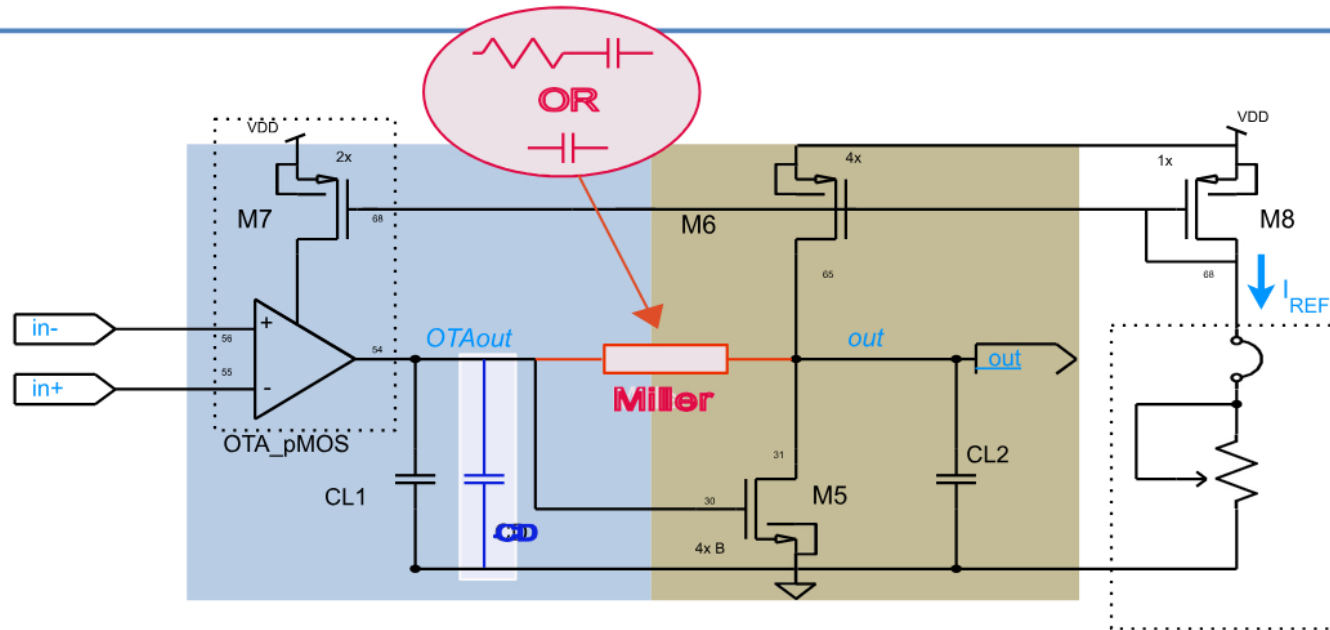
```
{ "1": [48, 50],  
  "2": [31, 49, 52, 53, 59],  
  "3": [28, 30, 51, 58, 60],  
  "4": [29, 57]}
```

Bitstream



```
pip3 install -i https://test.pypi.org/simple MOSbiusTools
```

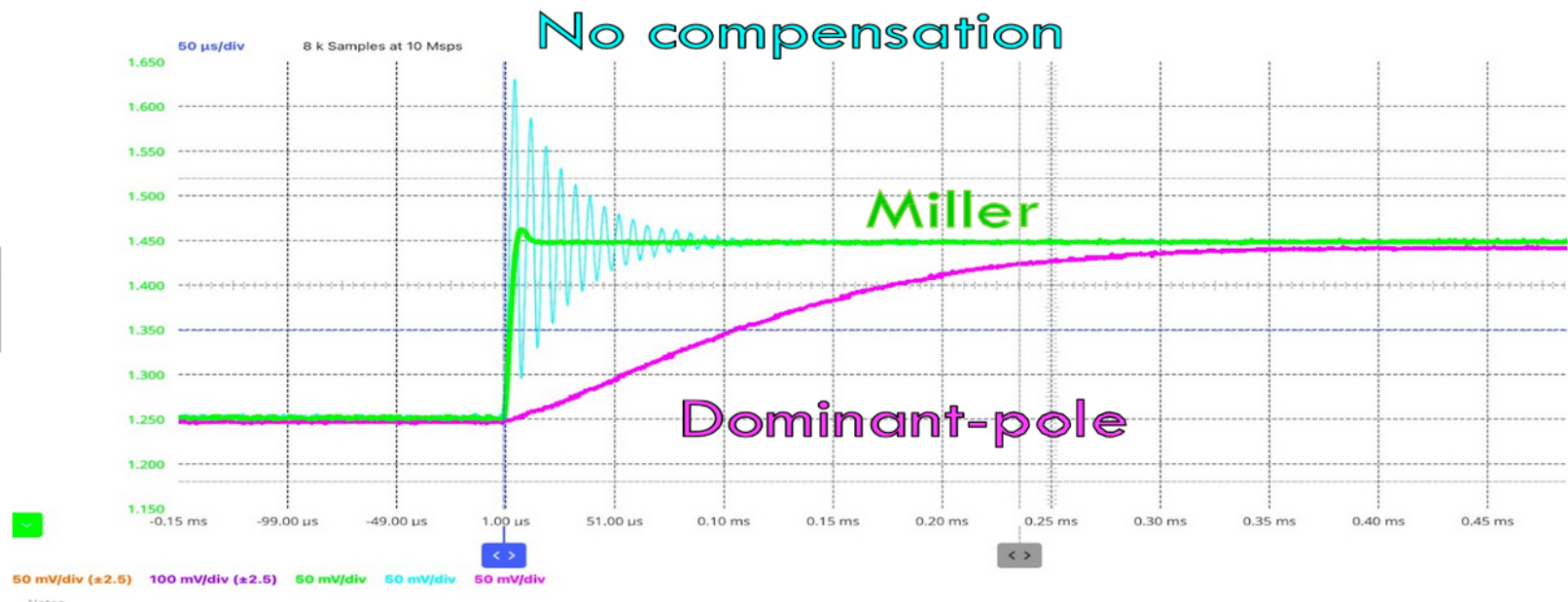
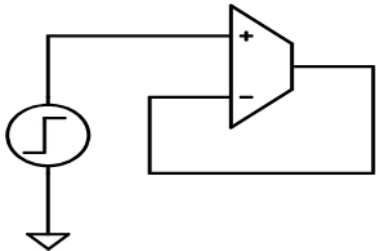
Lab Example: Compensate Two-Stage OTA



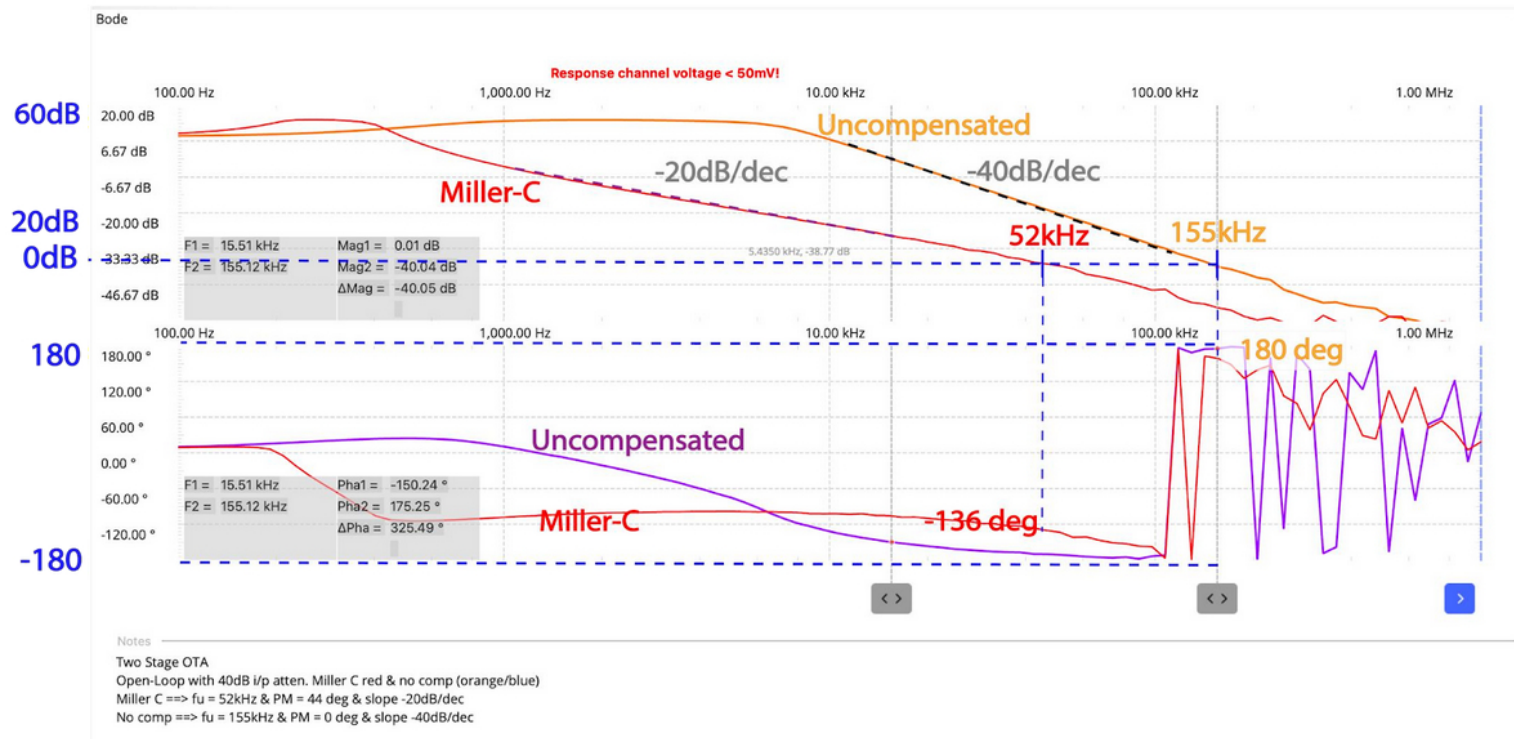
Dominant-Pole Compensation CD OR

Miller Compensation: C_c & R_c - C_c

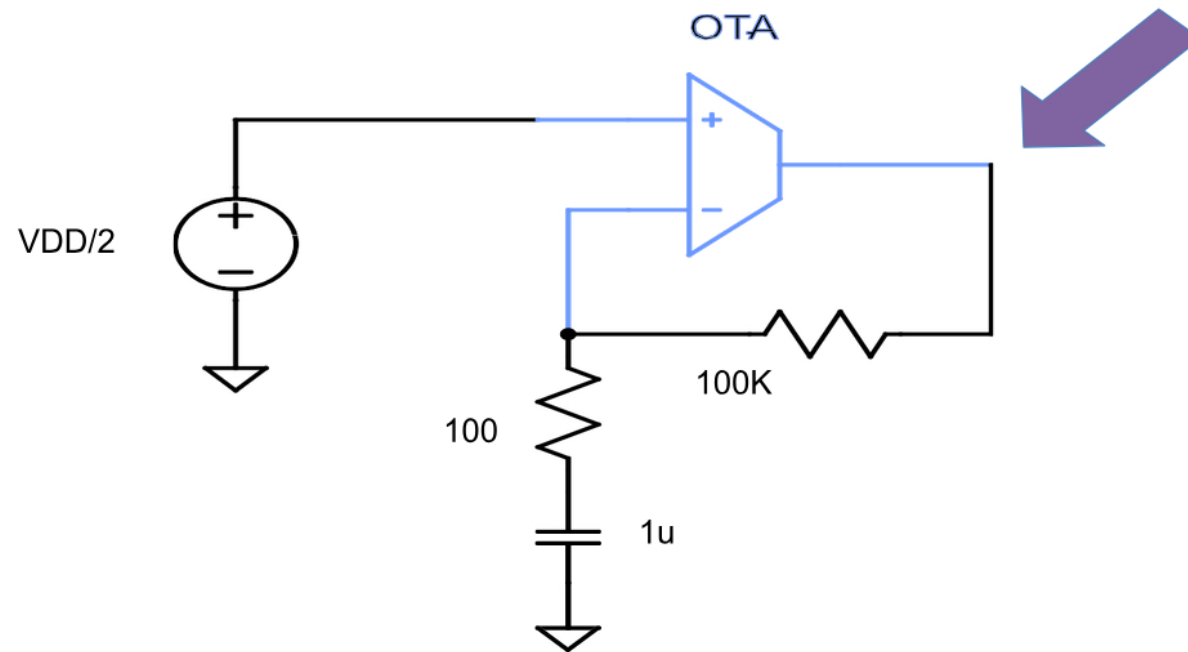
Unity-Gain Step Response



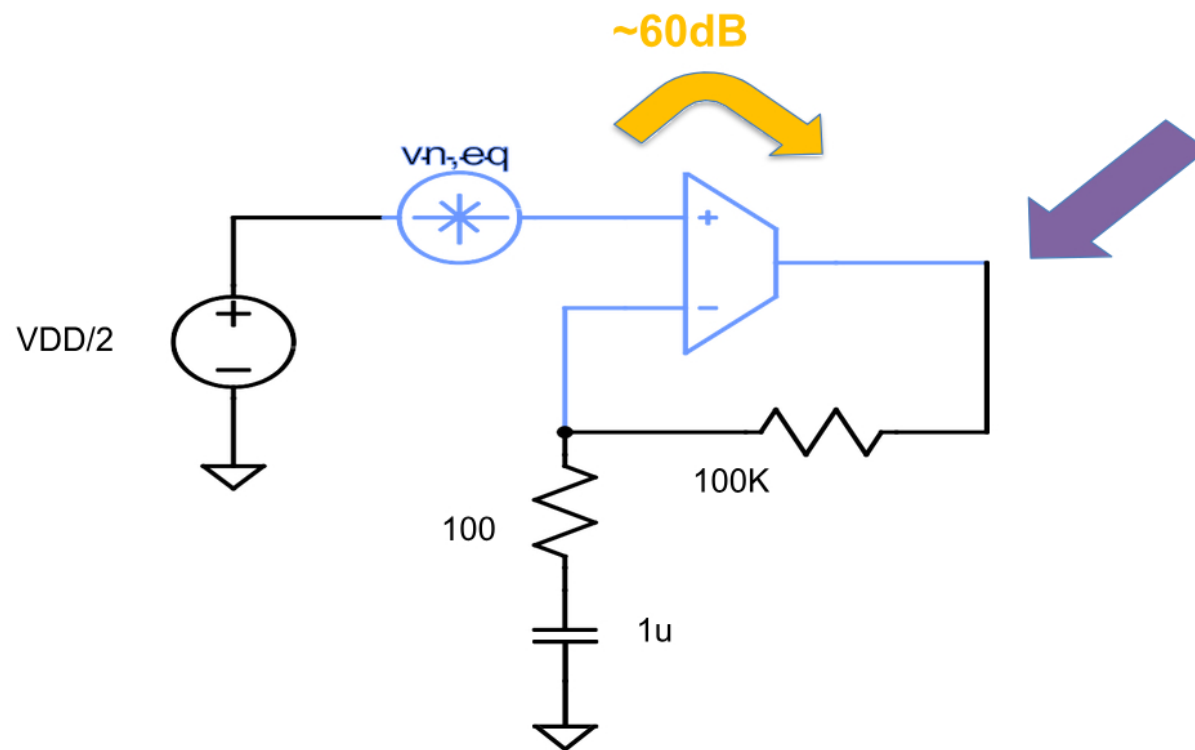
Loop Response: No C vs Miller-C



OTA in AC Open-Loop



OTA in AC Open-Loop



Video Interview with Matt Venn



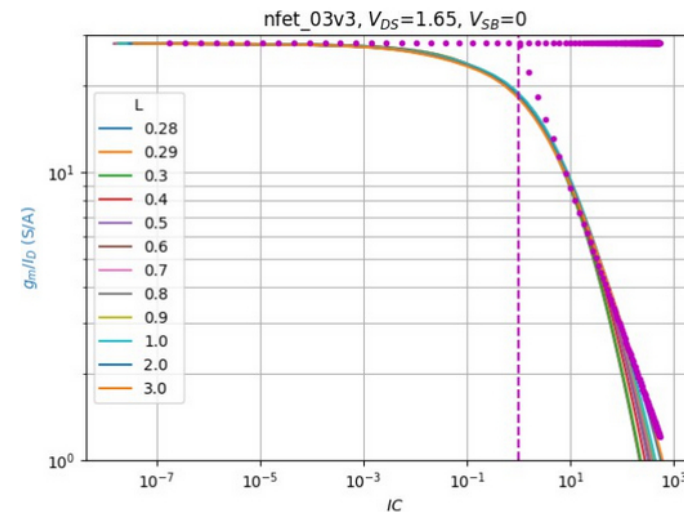
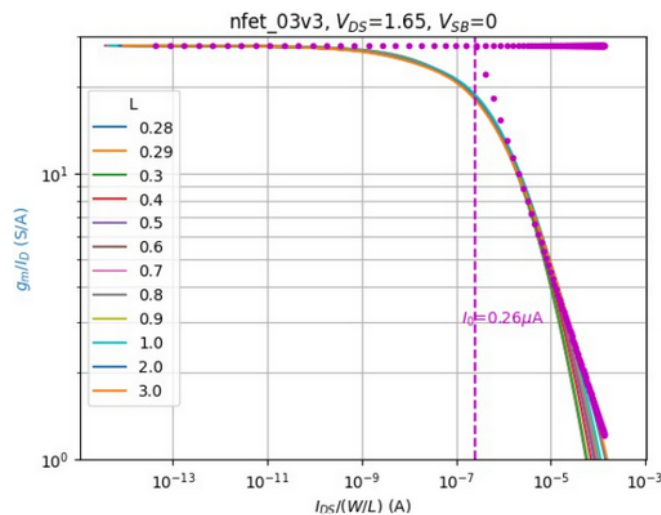
- Zero to ASIC Course
- June 25, 2025
- [MOSbius - A field programmable transistor array for chip designers - interview with Peter Kinget](#)
 - [One-Bit Delta-Sigma Converter Demo](#)

Transistor Bias Point Selection



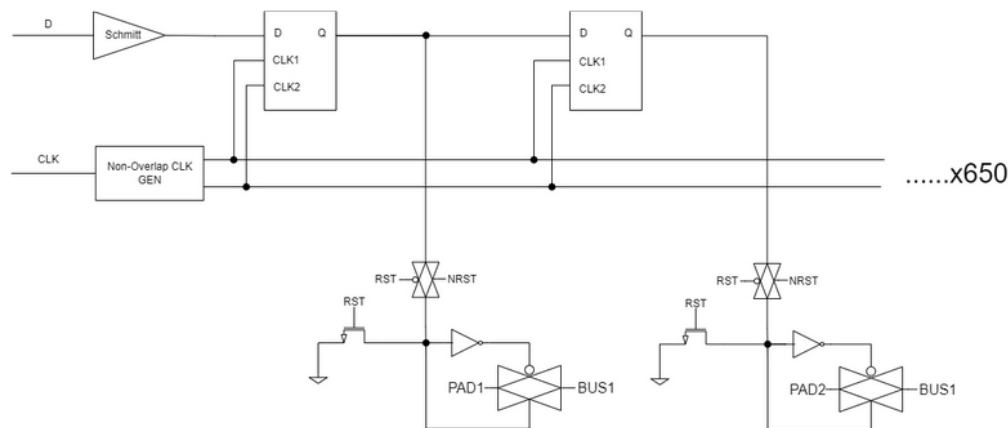
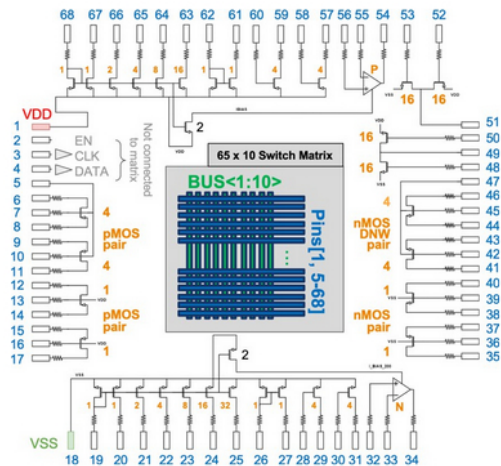
- Used a g_m/I_D approach and found the W/L so that:
 - $I_C = 10$ & $I_{DS} = 100\mu A$ & $V_{OV} \approx 200mV$ & $(g_m/I) \approx 9$
- In GF180, for 3.3V nMOS
 - $I_0 = 0.26\mu A$ & $I_C = 10$ & $g_m/I \approx 10 S/A$
 - for $I_{DS} = 2.6\mu A$ & $W/L = 1$,
 - for $I_{DS} = 26\mu A$ & $W/L = 10$, etc.

- Double Check yourself!!
- What is it for PMOS??
- Choose your own bias point



https://github.com/mosbiuschip/chipathon2025/blob/main/gf180mcuD_technology_info/README.md

- R_{on} of the Transmission Gate
 - 50-100 Ohm depending on your choice of bias point & transistor sizes
- Control Path
 - Shift Register $\Rightarrow$ careful with clocking (hold time violations ...) $\Rightarrow$ we generate non-overlapping clocks



Details for current MOSbius

How to architect your MOSbius chip:



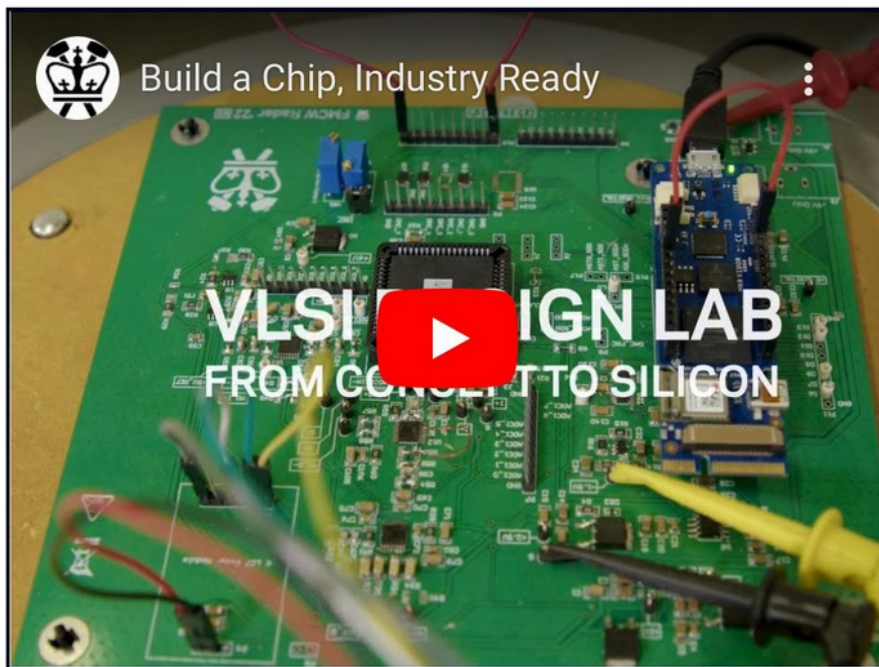
- Think about use cases for your chip
 - What experiments can your chip support?
 - How easy will your chip be to use, program, ...
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The design goals are not only about the design and tape-out, but also about your **chip's application**, i.e., what **experiments** your chip can support.

MOSbius Chipathon Ideas



- Port current ‘*analog*’ MOSbius (in part) to GF180
 - Significant benefits that everything will be in open-source, in particular, transistor models and layouts → better simulations, 3D animations, ...
- Develop ‘*digital-transistor-level*’ MOSbius
 - DFFs, logic gates, ... at circuit level
- (very advanced) RF MOSbius
 - RF building blocks to configure various RF receiver topologies
- **Your Ideas**



Many **detailed** chip designs: Class-D Amplifiers, PPGs, ECG, Sensor I/Fs, AM Receiver, WWVB Atomic Clock, FM Receiver, SAR ADCs, Digital Clocks, FPGA, RISC-V Processor, NPU, Ultrasonic Rangers, ...

Make a Chip Proposal



- Datasheet
 - Functionality of your chip & target specs
 - Block diagram
 - Functionality of building blocks
 - Schematics, if available, and design complexity of building blocks
 - Pin-out
 - Application diagrams
 - How do you test/demonstrate your chip
- Team members
 - Work distribution across team
- Schedule
 - Week by week
 - Align with Chipathon schedule

Put it in your GitHub project repository & share with mentors on discord so we can start reviewing and give feedback